

Phase 1: Requirements

Marklin WiFi AC Train Controller

2026-03-04

Test Project: Requirements Analysis

Design Input (raw)

A WiFi-enabled controller for classic Marklin AC model trains. The module sits inside (or replaces) the locomotive's existing electronics, is powered directly from the 24 VAC rail voltage, and exposes motor control via Modbus TCP over WiFi.

A simple and readily available microcontroller should be used, that is well supported and have good SW and IDE available.

Software development is not a part of this project, but the SW support for the chosen CPU/MCU must be considered.

Proposed Structured Requirements

1. Functional Requirements

- Control a Marklin AC motor with two field windings (one for forward, one for reverse)
- Direction control by selecting which winding is energized
- The two field windings must never be energized simultaneously. Hardware interlock (gate logic or equivalent) is required to enforce mutual exclusion independent of firmware
- WiFi connectivity for remote control
- Modbus TCP server for motor speed/direction commands
- Status feedback (at minimum: connected/running indication)

2. Power Input

- Source: 24 VAC nominal from Marklin track (range ~16-25 VAC depending on transformer)
- Must rectify and regulate to logic-level DC (3.3V for MCU, potentially intermediate rail ~12-15V)
- No external DC supply - all power from track

3. Interfaces

- Track input: 2-wire AC from rail (power + signal)
- Motor output: Connections to two field windings (forward and reverse) plus common
- WiFi: 802.11 b/g/n for Modbus TCP
- Status LED: At least one visible indicator
- Programming/Debug: USB or UART header for firmware upload (if MCU supports it natively)

4. MCU Selection Criteria

- WiFi built-in (reduces BOM, simplifies design)
- Well-supported toolchain and IDE (Arduino, ESP-IDF, PlatformIO)
- Sufficient GPIO for: motor drive signals, ADC for voltage sensing, LED, UART

- Low cost, readily available
- Small package suitable for locomotive interior
- Strong candidates: ESP32-S3, ESP32-C3, ESP32-C6

5. Environmental/Mechanical Constraints

- Must fit inside a Marklin locomotive body (compact PCB)
- Operating temperature: ambient (indoor model railway)
- No specific IP rating needed
- Components shall as far as possible be SMT; passives in 0603 packages

6. Applicable Standards

- IEC 62368-1:2023 (Ed. 4.0) - Audio/video, information and communication technology equipment - Part 1: Safety requirements
 - Hazard-based product safety standard that classifies energy sources, prescribes safeguards, and aims to reduce the likelihood of pain, injury, and property damage from fire.
 - Applies to electrical/electronic equipment in the audio, video, ICT, and office machine fields with rated voltage not exceeding 600 V.
 - Replaces the former IEC 60065 (AV safety) and IEC 60950-1 (IT equipment safety).
 - Reference: <https://webstore.iec.ch/en/publication/69308>

7. Out of Scope

- Firmware/software development
- Mechanical enclosure design
- DCC protocol support (this is for classic AC Marklin only)